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**Jain et al.**

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(54) **LINK LOSS DETECTION**

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*H04L 1/006* (2013.01); *H04M 11/062*

(2013.01)

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*H04L 1/0061*

See application file for complete search history.

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U.S.C. 154(b) by 750 days.

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*Primary Examiner* — Parth Patel

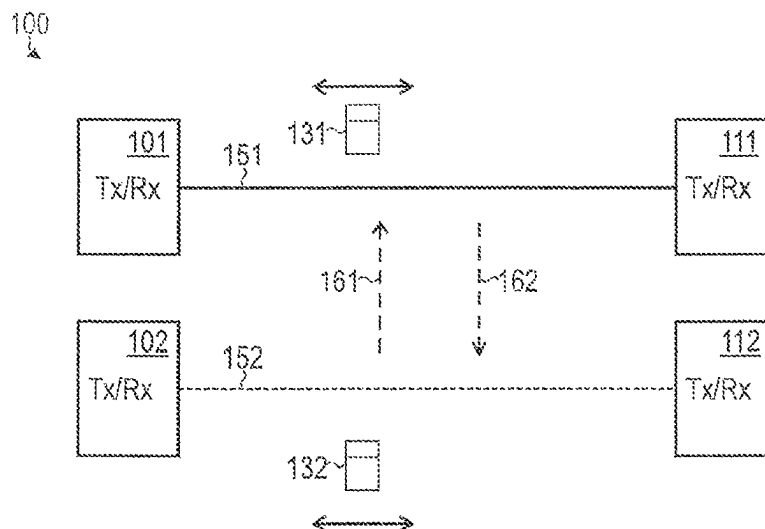
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**ABSTRACT**

A coded signal is received via a physical link and decoded.  
A link loss of the physical link is detected based on at least  
one of the coded signal and said decoding.

**20 Claims, 8 Drawing Sheets**



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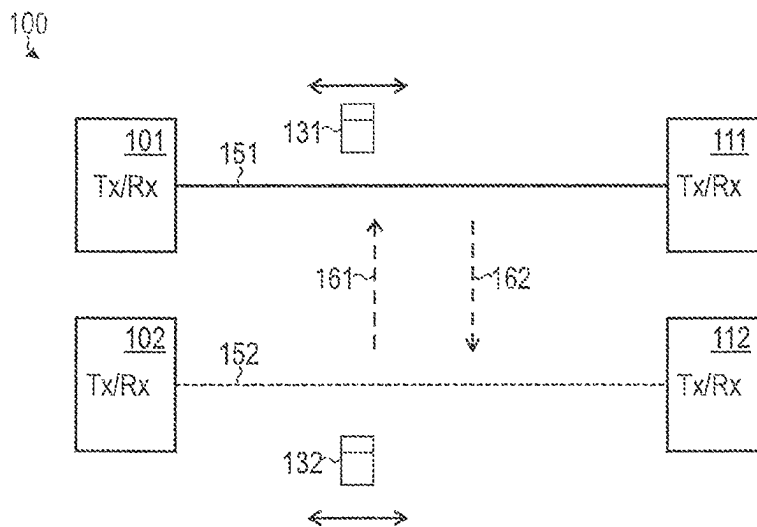


FIG. 1

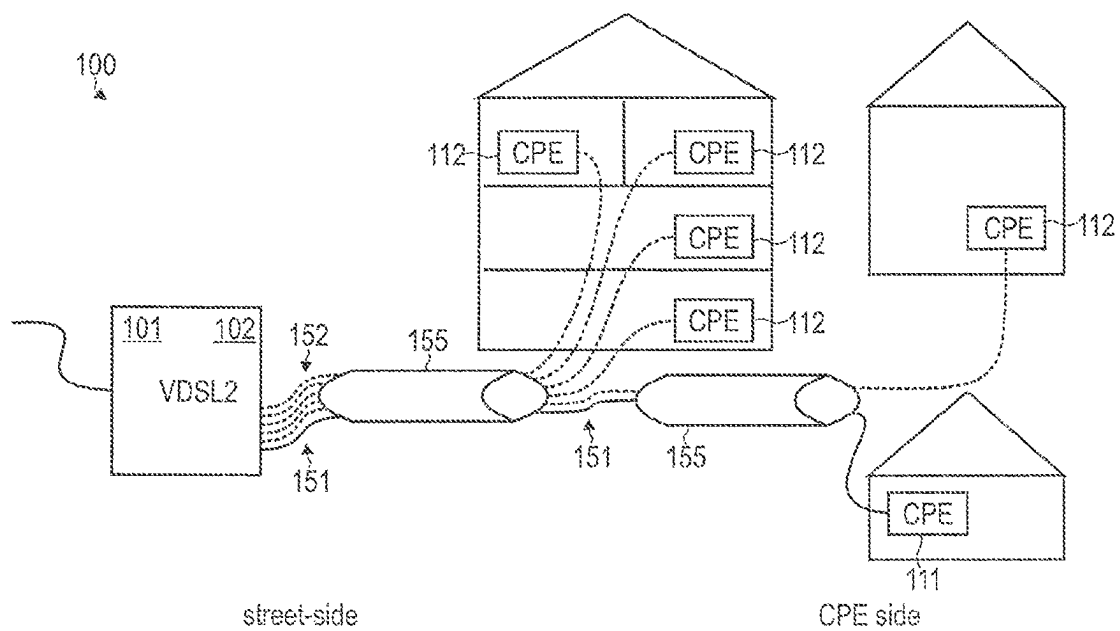


FIG. 2

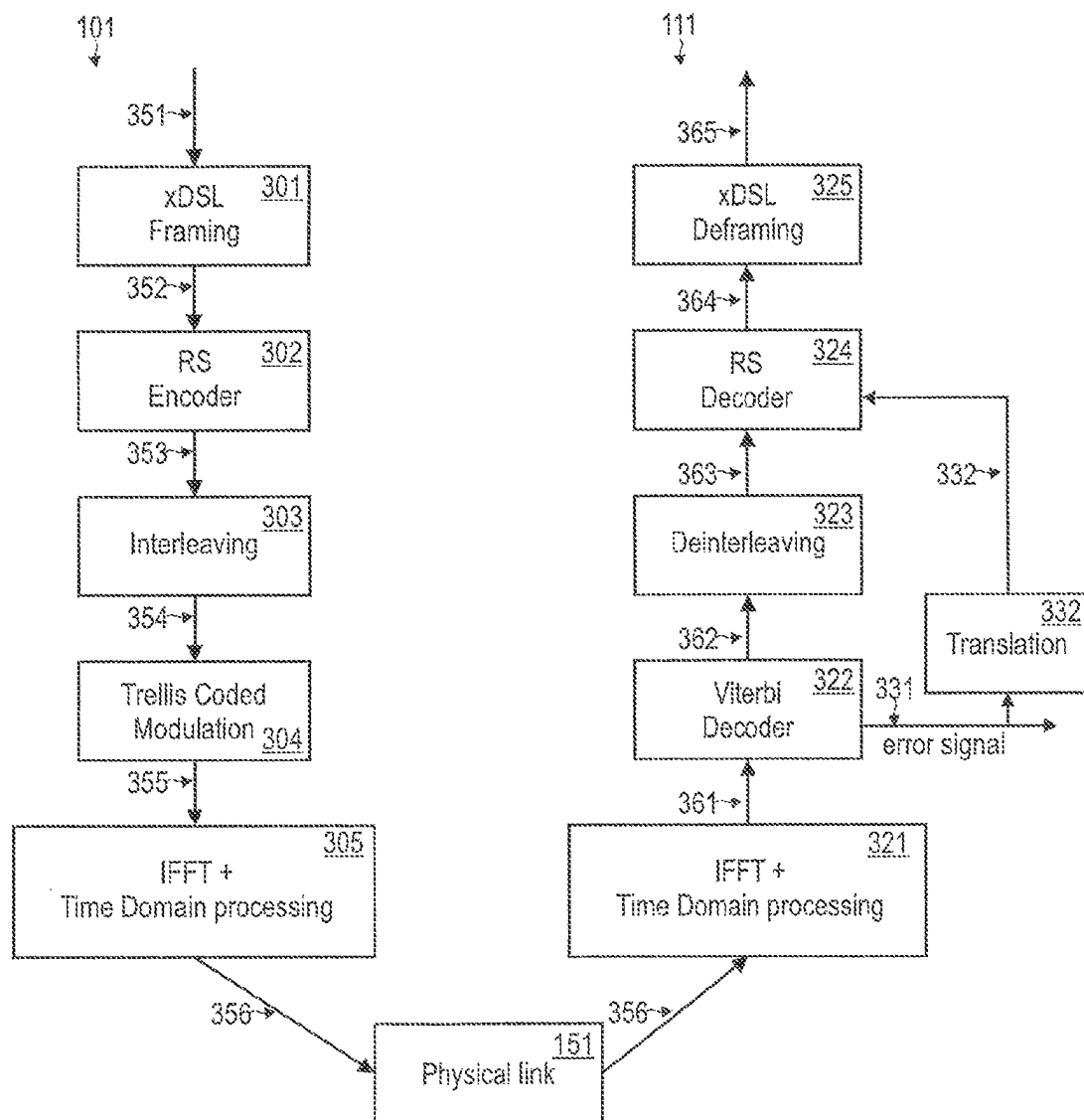


FIG. 3A

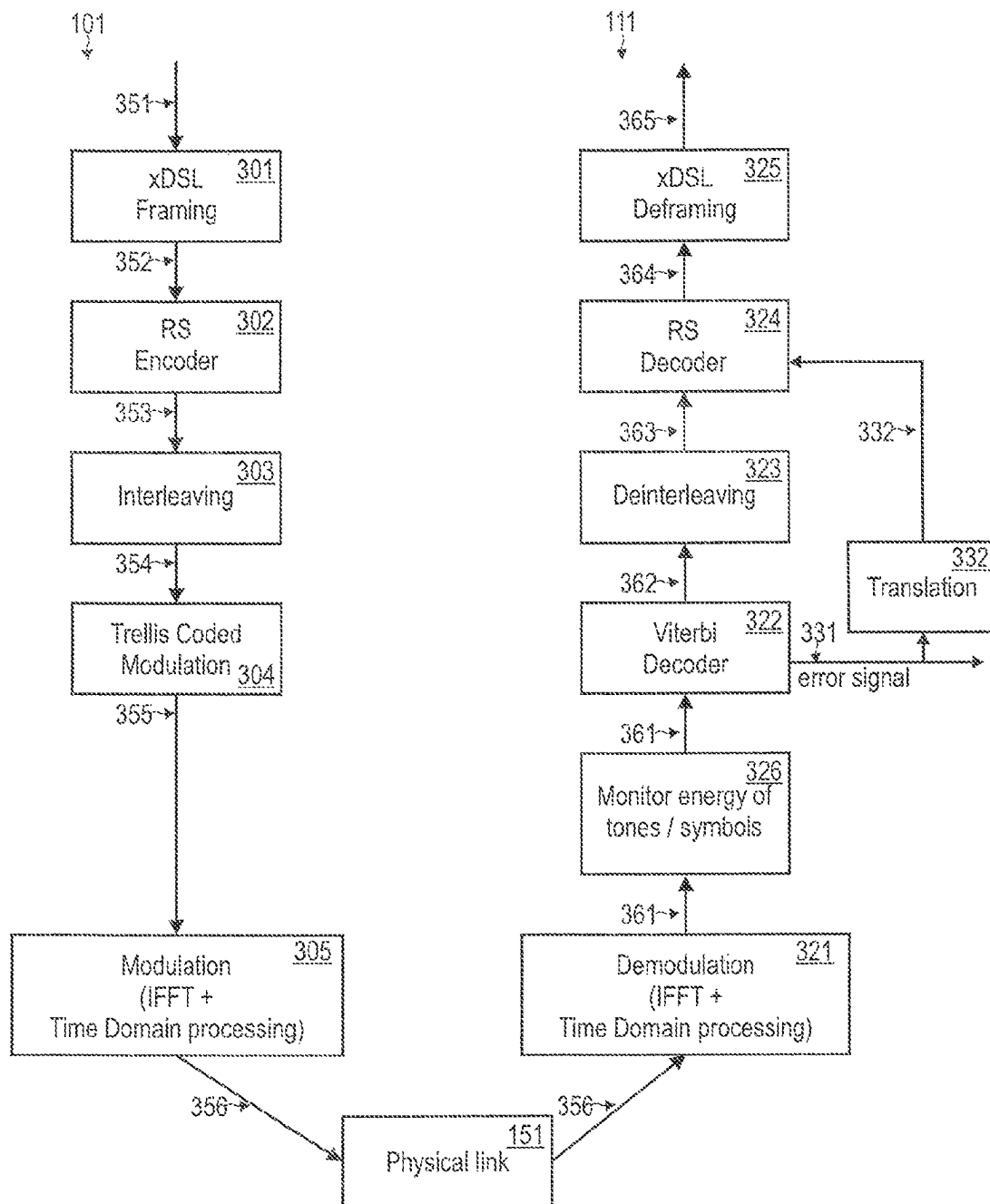


FIG. 3B

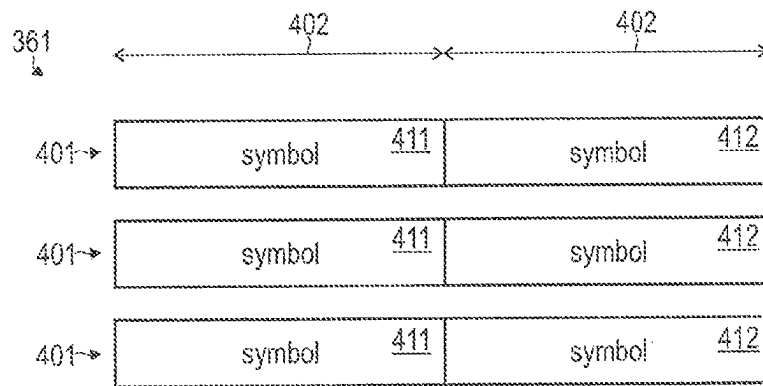


FIG. 4A

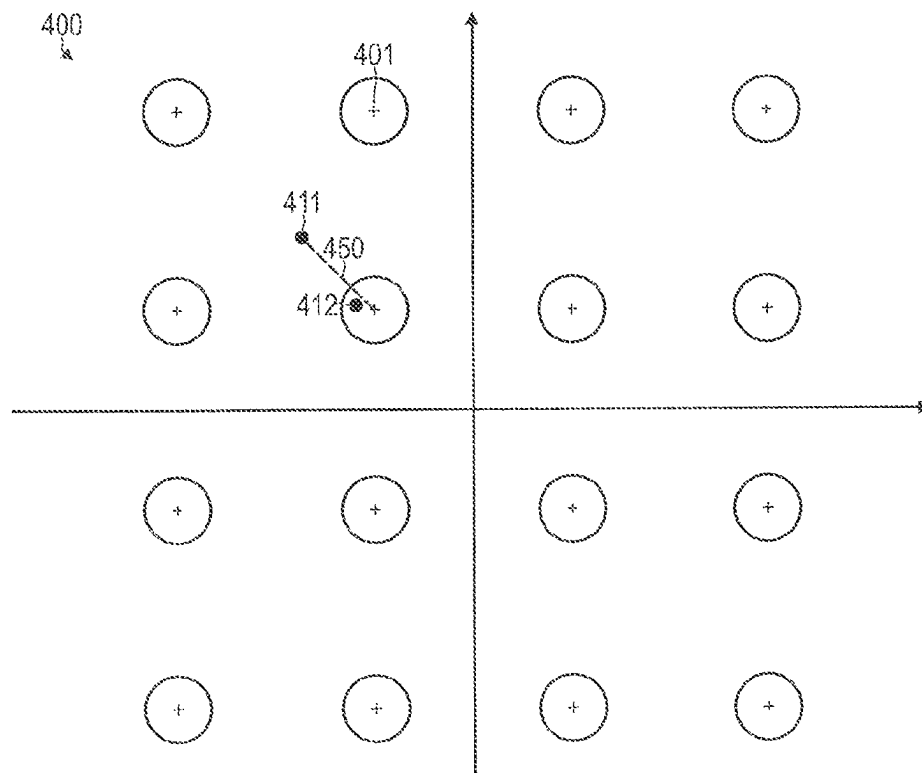


FIG. 4B

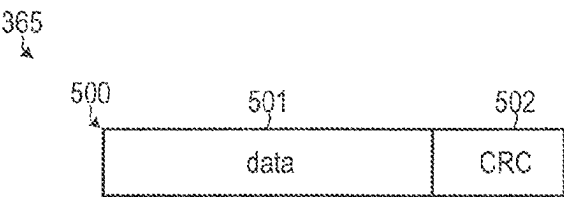


FIG. 5

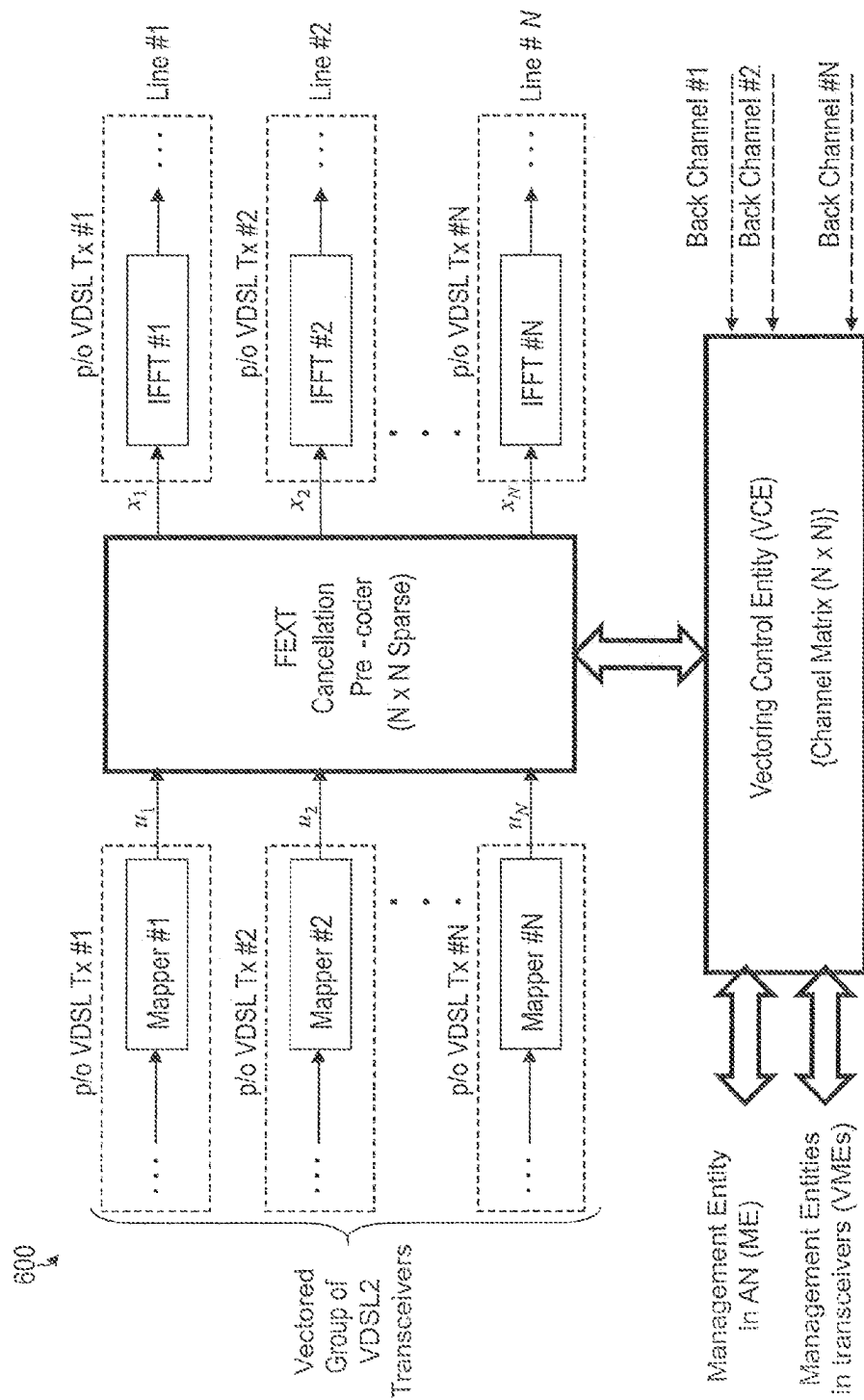


FIG. 6



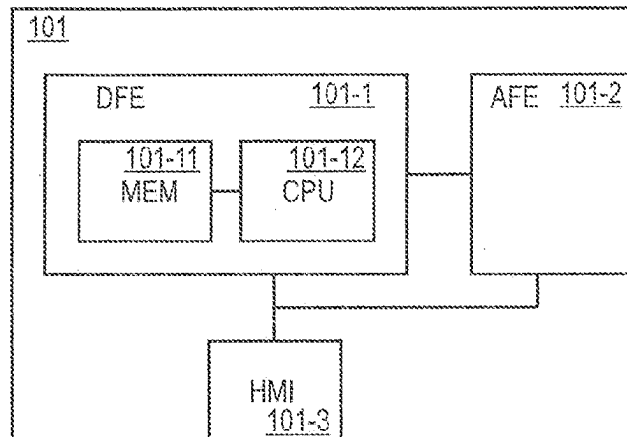


FIG. 7

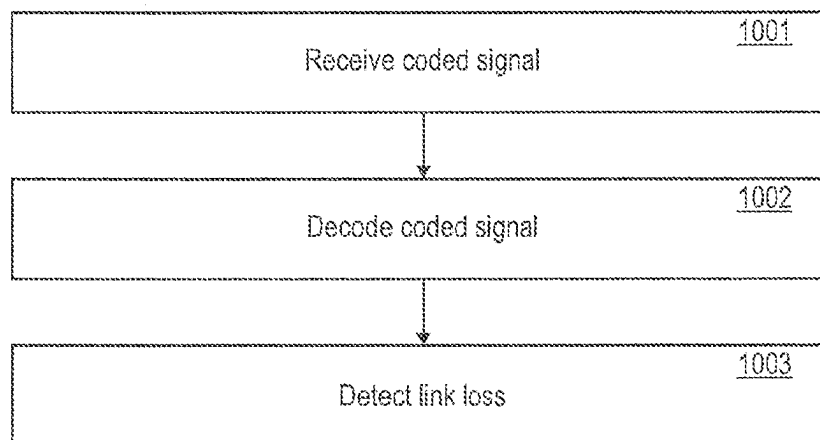


FIG. 8

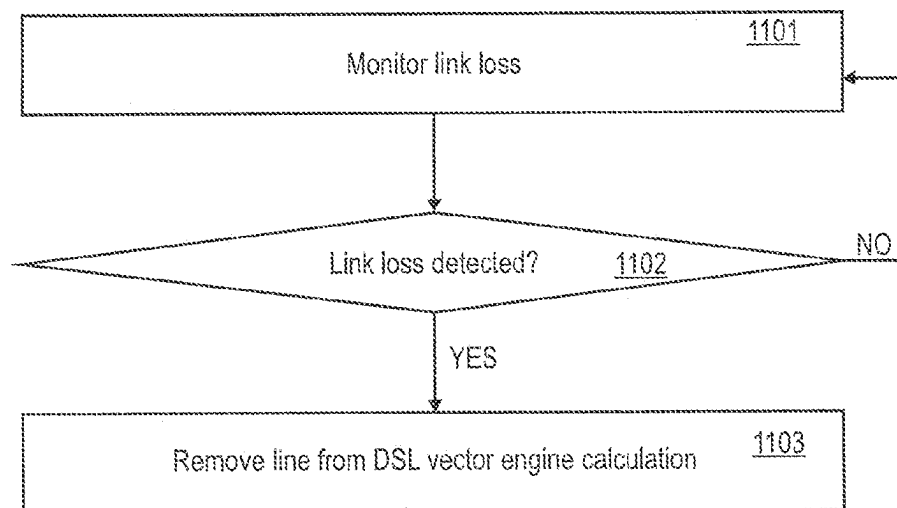


FIG. 9

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**LINK LOSS DETECTION**

This application is a National Phase entry application of International Patent Application No. PCT/EP2015/076965 filed on Nov. 18, 2015 which claims priority to U.S. Provisional Application 62/081,592 filed on Nov. 19, 2014 the contents of which are herein incorporated by reference in their entirety.

**FIELD**

Various embodiments relate to a method comprising detecting a link loss and to a corresponding device. In particular, various techniques relate to detecting a link loss of a physical link based on at least one of a coded signal and decoding of the coded signal.

**BACKGROUND**

Detecting the permanent removal of a physical link (link loss) can be helpful for controlling communication in a communication system. E.g., in context of Digital Subscriber Line (DSL) communication systems employing vectoring for removal of far-end crosstalk (FEXT), removal of physical links subject to link loss from a DSL vector engine calculation can be important in order to avoid negative impacts on remaining DSL channels handled by the DSL vector engine calculation.

Reference implementations for detecting link loss of a physical link typically detect the link loss with a comparably large latency and act slowly. E.g., the latency can be as high as between 2 and 10 seconds, e.g., according to the International Telecommunications Union (ITU) Telecommunication Standardization Sector (ITU-T) G.993.2 (2006), section 12.1.4.

Such a comparably high latency of detecting link loss can impose significant challenges on DSL vector engine calculations. When a line leaves, it can take a significant amount of time to detect the disconnection of the line. This typically results in a performance loss of data rate during this transitioning phase between link loss and detection of link loss.

**SUMMARY**

Therefore, a need exists for advanced techniques of detecting link loss of a physical link. In particular, a need exists for techniques which enable detecting the link loss at a comparably low latency and with a comparably high accuracy.

This need is met by the features of the independent claims. The features of the dependent claims define embodiments.

According to various embodiments, a method comprises receiving a coded signal via a physical link. The method further comprises decoding the coded signal to obtain a result signal. The method further comprises detecting a link loss of the physical link based on at least one of the coded signal and said decoding of the coded signal.

According to various embodiments, a device is provided. The device comprises a receiver configured to receive a coded signal via a physical link. The device further comprises a decoder configured to decode the coded signal to obtain a result signal. The device further comprises at least one processor configured to detect a link loss of the physical link based on at least one of the coded signal and said decoding of the coded signal.

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According to various embodiments, a computer program product is provided. The computer program product comprises program code to be executed by at least one processor. Executing the program code causes the at least one processor to execute a method. The method comprises receiving a coded signal via a physical link. The method further comprises decoding the coded signal to obtain a result signal. The method further comprises detecting a link loss of the physical link based on at least one of the coded signal and said decoding of the coded signal.

It is to be understood that the features mentioned above and those yet to be explained below may be used not only in the respective combinations indicated, but also in other combinations or in isolation without departing from the scope of the invention.

**BRIEF DESCRIPTION OF THE DRAWINGS**

In the following, the invention will be explained in further detail with respect to embodiments illustrated in the accompanying drawings.

FIG. 1 schematically illustrates a communication system according to various embodiments, wherein the communication system comprises a physical link between two transceivers which experiences FEXT and near-end crosstalk (NEXT) from a further physical link arranged in the vicinity of the physical link.

FIG. 2 schematically illustrates a deployment scenario of a communication system according to various embodiments for implementing a DSL channel via a physical link being formed by a copper wire.

FIG. 3A schematically illustrates coding and modulation of a signal at a transmitter and decoding and demodulation of the signal at a corresponding receiver according to various embodiments.

FIG. 3B schematically illustrates coding and modulation of a signal at a transmitter and decoding and demodulation of the signal at a corresponding receiver according to various embodiments.

FIG. 4A illustrates resource blocks of a coded signal in time domain and frequency domain, the resource blocks being communicated between a transmitter and a receiver via a physical link according to various embodiments, the resource blocks corresponding to symbols.

FIG. 4B schematically illustrates a constellation diagram of decoding and demodulating a symbol according to various embodiments.

FIG. 5 illustrates a frame of a signal communicated between a transmitter and a receiver via a physical link according to various embodiments.

FIG. 6 illustrates a DSL vector engine calculation according to various embodiments.

FIG. 7 illustrates a device according to various embodiments.

FIG. 8 is a flowchart of a method according to various embodiments.

FIG. 9 is a flowchart of a method according to various embodiments.

**DETAILED DESCRIPTION OF EMBODIMENTS**

In the following, embodiments of the invention will be described in detail with reference to the accompanying drawings. It is to be understood that the following description of embodiments is not to be taken in a limiting sense. The scope of the invention is not intended to be limited by

the embodiments described hereinafter or by the drawings, which are taken to be illustrative only.

The drawings are to be regarded as being schematic representations and elements illustrated in the drawings are not necessarily shown to scale. Rather, the various elements are represented such that their function and general purpose become apparent to a person skilled in the art. Any connection or coupling between functional blocks, devices, components, or other physical or functional units shown in the drawings or described herein may also be implemented by an indirect connection or coupling. A coupling between components may also be established over a wireless connection. Functional blocks may be implemented in hardware, firmware, software, or a combination thereof.

Hereinafter, various techniques with respect to detecting a link loss of a physical link of a communication system are disclosed. Link loss may correspond to a scenario where the physical link is interrupted and communication is thus prohibited.

In some examples, the link loss of the physical link can be detected based on a coded signal which is communicated via the physical link. E.g., the link loss can be detected based on an energy of the coded signal for various sample points in time domain and/or for various sample points in frequency domain.

In further examples, the link loss of the physical link can be detected, alternatively or additionally, based on decoding of the coded signal. Depending on the particular communication system implementing the techniques disclosed herein, techniques employed in the context of said decoding of the coded signal can vary. E.g., for various decoders such as a Viterbi decoder, a low-density parity check (LDPC) decoder, a Quadrature Amplitude Modulation (QAM) decoder, etc., it becomes possible to provide an error signal indicative of the presence of errors in said decoding. Different error metrics can be employed to determine the error signal, e.g., depending on the particular type of the decoder.

By such techniques as illustrated above, it becomes possible to detect the link loss of the physical link comparably quickly, i.e., with a comparably low latency. This enables to take actions as appropriate shortly after occurrence of the link loss. These actions can relate to the physical link for which the link loss is detected; and/or can relate to further physical links which are in some way or the other affected by the link loss of the physical link. Depending on the particular communication system employed, a wide variety of actions is conceivable.

The techniques disclosed herein are generally applicable for various communication systems. Examples include such communication systems which communicate a coded signal according to Discrete Multitone (DMT) coding and modulation and/or Orthogonal Frequency Division Multiplexing (OFDM) coding and modulation. Examples include the Institute of Electrical and Electronics Engineers (IEEE) 802.11 Wireless Local Area Network (WLAN) communication protocol and the Third Generation Partnership Project (3GPP) Long-Term Evolution (LTE) or Universal Mobile Telecommunications system (UMTS) protocol. Further examples include Bluetooth and satellite communication. For illustrative purposes only, hereinafter, a particular focus will be put on physical links that are implemented via a copper wire and establish communication channels according to the DSL protocol. The DSL protocols include ITU-T G.992.X (ADSL and ADSL 2+), G.993.1 (VDSL1), G.993.2 (VDSL2), and G.9700/G.9701 (G.Fast).

E.g., the various techniques disclosed herein can be applicable for communication system employed for the

Internet of Things (IoT) where a large number of devices communicates. Here, a low latency for link loss detection can be advantageous to ensure robust signaling.

Making reference to FIG. 1, data **131** is sent and/or received (communicated) via a first physical link **151** and second data **132** is communicated via a second physical link **152**.

The first data **131** and second data **132** may be control data, higher-layer payload data, and/or training data. Techniques disclosed herein generally relate to uni-directional and/or bidirectional communication, e.g., upstream (US) and/or downstream (DS) communication. Depending on US or DS communication, corresponding transceivers **101**, **111**, **102**, **112** may operate as transmitters or receivers. Communicating via the physical links **151**, **152** may be according to a frequency-division duplexing scheme (FDD) or according to a time-division duplexing scheme (TDD).

The first and second physical links **151**, **152** experience mutual crosstalk, i.e., the first physical link **151** (second physical link **152**) experiences first crosstalk **161** (second crosstalk **162**) from the second physical link **152** (first physical link **151**). Sometimes, this mutual crosstalk is also referred to as alien crosstalk. The crosstalk **161**, **162** may comprise FEXT and/or NEXT.

The first and second physical links **151**, **152** also experience intrinsic crosstalk. So-called impulse noise may hit a specific physical link **151**, **152**. In the various examples disclosed herein, the link loss of the physical link **151** may be detected based on the temporal evolution of the received coded signal in a time interval having a duration in the range of 3-50 milliseconds, preferably in the range of 5-8 milliseconds. Monitoring for, e.g., erroneous symbols over such a time interval may be motivated by the finding that a typical duration of impulse noise, e.g., where the physical link **151** is implemented via copper wire, is of the duration of 2 to 5 milliseconds.

FIG. 2 illustrates aspects of a typical VDSL2 deployment scenario. The physical links **151**, **152** implemented as copper wires connect the Digital Subscriber Line Access Multiplexer (DSLAM) **101**, **102** to the VDSL2 transceiver units, sometimes referred to as remote terminals, at physical separated individual residences comprising customer premises equipment (CPE) **111**, **112**. The physical links **151**, **152** share common cable binders **155** which increases NEXT and FEXT. VDSL2 employs DMT modulation with up to a 4096 subcarriers located on frequencies spaced by 4.3125 kilohertz or 8.625 Kilohertz. Due to the multiple physical links **151**, **152** connected to the DSLAM **101**, **102** and sharing a cable binder **155**, NEXT and FEXT can be prominent.

Typically, NEXT is prominent above 1-2 MHz. Because of this, VDSL2 communication channels use non-overlapping DS/US frequency bands in FDD up to 30 MHz. This significantly mitigates NEXT. Thus, with NEXT being largely eliminated due to the FDD, FEXT typically dominates the remaining noise **161**, **162**. Crosstalk cancellation—also referred to as vector engine calculation for removing FEXT—significantly reduces the FEXT, thus effecting a performance improvement.

The vector engine calculation for removing FEXT on the physical link **152** should have access to information whether a given physical link **151** is connected or disconnected. Because of this, it can be helpful to detect the link loss of one of the physical links **151** at a comparably low latency. In detail, in a communication system as illustrated in FIGS. 1 and 2, each physical link **152** typically maintains its specific crosstalk coefficient or pre-coder coefficient corresponding to neighboring physical links **151**. Each crosstalk coefficient

of the particular line **152** (victim physical link) is estimated during a training, E.g., by measuring the effect of each neighboring physical link (disturber line) **151** on the victim physical link **152**. Employing the DSL vector engine calculation, data communicated on a victim physical link **152** is manipulated by using crosstalk or pre-coder coefficients at the transmitter **101**, **102**, **111**, **112** such that crosstalk **161**, **162** of disturber physical links **151** is reduced. During a transitioning phase after link loss of the disturber physical link **151**, data communicated on the victim physical link **152** is still manipulated by using crosstalk or pre-coder coefficients having been determined before the link loss. Therefore, the data communicated on the victim physical link **152** is artificially manipulated as if the disturber physical link **151** was actually still active. This reduces reliability of communication on the victim physical link **152**. Therefore, it can be desirable to detect the link loss of the disturber physical link **151** comparably fast. Hereinafter, techniques are disclosed which enable to detect the link loss of the disturber physical link **151** comparably fast.

Link loss of the physical link **151** may occur where the corresponding copper wire is physically broken. A further source for link loss may be malfunctioning of the CPE **111** associated with the physical link **151**. E.g., the CPE **111** may be powered down.

Techniques are disclosed herein which enable to reliably detect link loss of a disturber physical link **151** at a low latency.

FIG. **3A** illustrates an embodiment which enables to detect the link loss of a physical link **151** comparably quickly. FIG. **3A** illustrates an OFDM-based communication system.

Here, combined Forward Error Correction (FEC) based on checksums comprised in transmission frames, time and/or frequency interleaving, and/or Viterbi encoding is used for combating the effects of impulse noise affecting the physical link **151**. The FEC is typically implemented by a redundancy encoder such as a LDPC decoder or a Reed-Solomon decoder. By providing an error signal indicative of the presence of errors in the signal output by the Viterbi decoder, a correction capability of the redundancy decoder can be almost doubled. Hereinafter, scenarios are disclosed, where the error signal is re-used for detecting link loss of the physical link **151**.

The signal transmitted via the physical link is encoded and modulated by the transmitter **101** and decoded and demodulated by the receiver **111**. For this, the OFDM-based communication system of FIG. **3A** employs a plurality of carriers or tones which act as separate communication channels implemented via the physical link **151** to carry information between the transmitter **101** and the receiver **111**. Each carrier is a group of one or more frequencies defined by a center frequency and a predefined bandwidth.

The physical link **151** is subject to various types of interference and noise. Interference and noise can corrupt the signal **356** received at the receiver **111** if compared to the signal **356** transmitted at the transmitter **101**. Some sources of interference and noise can be modeled as additive white caution noise (AWGN). The impact of AWGN can be reduced greatly by channel estimation and channel decoding employing a Viterbi decoder. Channel estimation typically computes the signal-to-noise ratio (SNR) of the received signal **256** at the receiver **111**. According to OFDM techniques, based on the computed SNR of each carrier, the number of data bits loaded on each carrier is determined (bit loading). Lower bit loading typically improves robustness of communication against errors.

Now explaining the functioning of the OFDM-based communication system of FIG. **3A** in detail, at the transmitter **101**, packetized data **351** is mapped to transmission frames at framing **301**. The data **352** is then encoded by, e.g., RS encoding **302**, to implement FEC. An interleaver **303** interleaves the encoded data **353**, e.g., in time domain, to increase a robustness against impulse noise. The interleaved data **354** is then further encoded, e.g., using a Trellis coded modulation (TCM) encoder **304** or a QAM encoder (the latter not shown in FIG. **3A**). The encoding at **304** further modulates the signal **354** onto different carriers of a DMT. Time and frequency domain processing is then performed at **305**, e.g., comprising further interleaving and/or modulation onto different carriers in the high frequency spectrum and/or digital-to-analog conversion.

Thus, a coded signal **356** is communicated via the physical link **151** and received by a receiver **111**. First, the coded signal **356** is processed in time and frequency domain **321**; e.g., samples of the received analog signal are converted into digital domain. Further, data of different carrier frequencies can be separated by employing inverted Fast Fourier Transformation (IFFT). Thus, a coded signal **361** is obtained in digital domain.

An example structure of the signal **361** is illustrated in FIG. **4A**. The signal comprises a plurality of symbols **411**, **412**, each symbol occupying a certain time resource block **402** and frequency resource block **401**. The symbols **411**, **412** are referred to DMT symbols. The different symbols **411**, **412** may be separated in time domain by guard intervals (not shown in FIG. **4A**). The different carriers **411** may be separated in frequency domain and/or may carry different phases. Each of the symbols **411**, **412** may correspond to a sequence of bits comprising a number of bits as defined by the bit loading.

Referring again to FIG. **3A**, a decoder **322** then decodes the signal **361**. E.g., the decoder **322** can be a QAM decoder or a unit combining QAM decoding and Viterbi decoding. In the example of FIG. **3A**, a Viterbi decoder **322** is employed. The decoder **322** attempts to reconstruct the symbols input into the encoder **304** in view of potential corruption by noise on the physical link **151**. In case the encoder **304** uses QAM encoding, the decoder **322** also uses QAM decoding. In case the encoder **304** uses TCM encoding—which includes QAM encoding—the decoder **322** also uses QAM decoding, followed by Viterbi decoding.

The reconstructed symbols are output by the decoder **322** as signal **362** and are input to a deinterleaver **323**. The deinterleaver **323** produces the interleaved data as signal **363** which is provided to a second-stage decoder **324**, i.e., in the example of FIG. **3A**, a RS decoder **324**. The decoder **324** provides the finally decoded result signal **364** to a de-framing unit **325** which strips off the transmission frames to provide higher-layer packetized data **365**.

In the example of FIG. **3A**, the Viterbi decoder **322** outputs an error signal **331** indicative of a presence of errors in said decoding of the Viterbi decoder **322**. E.g., the error signal **331** can indicate which of the carrier symbols of signals **362** are likely to be erroneous. E.g., the decoder **322** may indicate that a whole DMT symbol **411**, **412** of the signal **362** is corrupt. Based on interleaving properties, a translation unit **332** processes the addresses of the bits in the corrupt DMT symbols **411**, **412** to form address data which indicates the addresses of bits in the de-interleaved signal **363**. This address data is input to the RS decoder **324** so that the Reed Solomon decoder **324** is enabled to perform, e.g., erasure decoding. As can be seen from the above, the RS decoder **324** operates as a second-stage redundancy decoder

based on the error signal **331**. By providing redundancy coding/decoding, a likelihood of errors in the signal **364** can be further reduced. Corresponding techniques are described in detail in U.S. Pat. No. 7,743,313 B2, the entire disclosure of which is incorporated herein by reference, such that further details are not required to be illustrated in the present context.

Hereinafter, details of determining the error signal **331** are explained. E.g., where—as in the scenario of FIG. 3A—a Viterbi decoder **322** is employed, the survival path, sometimes also referred to as Viterbi path, having an extreme value of the corresponding metric is selected for providing the result signal **362**. Typically, other Viterbi paths have significantly different metrics compared to the survival path in a scenario where Gaussian noise is present. This facilitates selection of the result signal **362**; in particular, a confidence in selecting the result signal **362** may be comparably high. However, where impulse noise impacts the communication via the physical link **151**, the metric values of all Viterbi paths are typically of the same order of magnitude for a substantial number of TCM encoding stages, i.e., a difference between different Viterbi paths of the Viterbi decoder is comparably small. In this case, selecting the survival path for providing the result signal **362** can become difficult and the confidence in selecting the result signal **362** may drop. Thus, in a scenario where the difference between different Viterbi paths of the Viterbi decoder is comparably small—e.g., below a predefined threshold—the respective symbol of the result signal **362** is marked as erroneous in the error signal **331**.

Above, example scenarios have been illustrated where the error signal **331** is provided by the Viterbi decoder **322**. However, different examples, different decoders may be employed, such as a QAM decoder and/or a LDPC decoder. Also in such scenarios, it is possible to determine the error signal **331**. Corresponding aspects are illustrated with respect to FIG. 4B.

FIG. 4B illustrates a constellation diagram **400** of, e.g., a QAM decoder or a LDPC decoder. The example of FIG. 4B in particular shows a 16-QAM constellation, in which each of 2 quadrature waves is modulated to take one of four possible amplitude values, so that the constellation includes 16 points in total. In different scenarios, different constellations may be employed. In particular, e.g., in DSL communication, different bit loading may be employed to encode a different number of bits per symbol **411**, **412**. As in conventional decoders, to obtain an estimate of what data the encoder **304** at the transmitter **101** intended to encode, the decoder **322** identifies which point of the corresponding constellation is closest to the received carrier symbol **401**. Here, different metrics may be employed such as the Euclidean distance—which is typically employed for QAM—or the log-likelihood estimate—which is typically employed at the LDPC decoder. The error signal **331** may be indicative of the distance between the decoded symbol **411**, **412** and the respective carrier **401**. E.g., the error signal **331** may indicate the distance in quantitative terms. In other examples, the error signal **331** may indicate the distance in qualitative terms, only. E.g., the error signal **331** may flag a respective symbol as potentially being erroneous if the distance **450** of the corresponding symbol **411**, **412** exceeds a certain threshold (illustrated by the circles in FIG. 4B).

Above, various techniques have been illustrated in order to provide and determine the error signal **331** indicative of the presence of errors in said decoding. Where the link loss is detected based on said decoding, it is now possible to employ the error signal **331** to identify the link loss. Here,

different techniques may be employed for detecting the link loss depending on the error signal **331**; in particular, the techniques may vary depending on the information content of the error signal **331**.

Referring again to FIG. 3A, based on the error signal **331** provided by said decoding **322**, it is then possible to detect the link loss. E.g., where the error signal **331** indicates a comparably high likelihood for a number of subsequent erroneous symbols **411**, **412**—e.g., corresponding to the above-mentioned time interval in the range of 3-15 ms—link loss may be detected. In one example, the error signal **331** may be indicative of a number of adjacent (in time domain) erroneous symbols **411**, **412** of the coded signal. Then, it is possible to execute a threshold comparison between the number of adjacent erroneous symbols **411**, **412** and a predefined threshold. The link loss may be detected based on said executing a threshold comparison. E.g., a counter may be maintained which is incremented for each continuous, erroneous symbols **411**, **412**. E.g., if an adjacent number of 5, 10, 50, or hundred symbols **411**, **412** is erroneous/corrupted, link loss may be detected. In particular, the number of adjacent erroneous symbols **411**, **412** may vary depending on properties such as a typical duration of impulse noise on the physical link **151** and/or a typical duration of the symbols **411**, **412** and/or bit loading.

While above techniques have been disclosed which detect the link loss based on the error signal **331** of the first-stage decoder **322**, in other examples the link loss may be alternatively or additionally detected based on the second-stage redundancy decoding **324**. In some examples the second-stage redundancy decoder may output a further error signal (not shown in FIG. 3A) which may be used to detect the link loss.

Now turning to FIG. 3B, aspects of detecting the link loss of the physical link **151** based on the coded signal **361** are disclosed. E.g., it is possible to monitor, at **326**, the energy of certain frequencies and/or time samples of the signal **361**. The frequencies may correspond to certain carriers of the coded signal **356**. The time samples may correspond to symbols of the signal **361**. However, monitoring the energy in time domain and/or frequency domain may also be done independently of the time and/or frequency spacing of the coded signal **356**.

In some examples, the energy levels of a plurality of resource blocks of the coded signal can be measured and the link loss can be determined based on the measured energy levels. Here, the resource blocks can correspond to the symbols and/or carriers of the coded signal **361**. Then, a threshold comparison between the energy levels of the plurality of resource blocks and a predetermined threshold can be executed. The link loss may be determined based on said executing of the threshold comparison.

In some examples, adjacent time samples of the coded signal **361** and/or adjacent carriers of the coded signal **361** are monitored at **326**. Thereby, the time evolution of the energy of the resource blocks can be tracked, thereby identifying the link loss more reliably. E.g., a counter may be maintained which is incremented for continuous, adjacent time samples and/or frequency samples of the coded signal **361** that have an energy below the predefined threshold. The counter may then compared to the predetermined threshold. If the number of adjacent time samples and/or frequency samples of the coded signal **361** is above a threshold, link loss may be detected. E.g., in time domain processing, if the energy of a certain amount of samples of the signal **361** is below the threshold over multiple symbols **411**, **412**, link loss can be detected. Typically, such a scenario equals a

situation where no data communication via the physical link **151** is happening and only background noise is picked up by the receiver **111** and the physical link **151**. Thus, link loss of the physical link **151** has occurred. E.g., in frequency domain processing, after IFFT de-modulation, if the energy of a certain amount of carriers in a specific frequency band of the signal **361** is below a certain threshold over multiple symbols **411**, **412**, the link loss can be detected. Again, the remaining energy picked up by the receiver **111** and the physical link **151** can be due to background noise.

FIG. **5** illustrates aspects with respect to the result signal **365**. The result signal **365** comprises the sequence of bits (not shown in FIG. **5**) forming transmission frames **500**. The transmission frames **500** comprise a data section **501** carrying higher-layer payload data in the checksum section **502** comprising a checksum such as a cyclic redundancy check (CRC). In some scenarios, it may be desirable to complement the detection of link loss of the physical link **151** implemented at early stages of the receiver **111**—as explained above with respect to FIGS. **3A** and **3B**—by detection of link loss implemented at a later stage of the receiver **111**. E.g., it may be possible that the link loss of the physical link **151** is further detected based on a checksum of at least one of the transmission frames **500** of the result signal **365**. Typically, such a detection of the link loss may be associated with a comparably high latency, but may be comparably accurate.

FIG. **6** illustrates details of a DSL vector engine calculation for removing FEXT between the physical link **151** and the plurality of further links **152**. Where link loss of the physical link **151** is detected, it is possible to remove the DSL channel implemented via the physical link **151** from the DSL vector engine calculation **600**, but retain the further DSL channels implemented via the further physical links **152** at the DSL vector engine calculation **600**. Where the transitioning phase is comparably small, because the link loss of the physical link **151** is detected at a low latency, negative impacts on the further DSL channels due the link loss can be mitigated.

FIG. **7** is a schematic illustration of a device **101** according to various embodiments. The device **101** implements a transceiver for communicating on the physical link **151**. The device implements communication on the physical link **151** via, e.g., a DSL channel. The device **101** comprises an analog front end (AFE) **101-2** and a digital front end (DFE) **101-1**. Typically, the AFE **101-2** implements time domain and frequency domain processing **321** of the raw coded signal **356** received via an antenna or the like. The DFE **101-1** comprises a processor **101-12** and a memory **101-11**. The memory **101-11** stores program code that may be executed by the processor **101-12** and may cause the processor **101-12** to execute techniques as illustrated above with respect to blocks **326**, **322**, **323**, **324**, and **325**. In particular, the processor **101-12** may be configured to demodulate and/or decode the digitized raw signal **361**. The device **101** further comprises a human machine interface (HMI) **101-3** configured to input information from a user and to output information to a user.

Executing program code stored at the memory **101-11** by the processor **101-12** may cause the processor **101-12** to execute the method as illustrated in FIG. **8**. First, at **1001**, the raw coded signal **356** is received. Potentially, frequency and/or time domain processing **321** is applied to the raw coded signal **356**, e.g., by the AFE **101-2**. Thereby, the coded signal **361** is obtained.

Next, at **1002**, the coded signal **361** is decoded, e.g., by a Viterbi decoder **321**, a QAM decoder, and/or a LDPC

decoder. Depending on the particular decoder employed, it may be required to provide an additional demodulation before decoding the coded signal **361**.

At **1003**, the link loss is detected. The link loss of the physical link **151** may be detected based on a temporal evolution of the coded signal **356**, **361** and/or said decoding; here, the temporal evolution may be considered for a duration in the range of 3-15 ms or 5-8 ms. In particular, the link loss may be detected based on properties of the decoding at **1002** and/or may be detected based on properties of the coded signal **361**, e.g., based on energy across a plurality of samples in time domain and/or frequency domain. Where the link loss is detected based on properties of the decoding at **1002**, the decoding may provide the error signal **331** based on which the link loss may be detected. It is also possible to detect the link loss based on second-stage decoding, e.g., by a RS decoder.

Steps **1001-1003** may be reiterated over the course of time in order to monitor the link loss, see FIG. **9**, **1101**. Once link loss is detected at **1102**, the corresponding physical line **151** can be removed from the DSL vector engine calculation **400**, **1103**. Because link loss can be detected comparably quickly, **1103** can be executed soon after the link loss actually occurred such that performance of further physical links **152** is not degraded for an extended transition phase.

Summarizing, above techniques have been disclosed which enable to detect loss of a physical link in a communication system with a comparably low latency. In particular, e.g. for an application within a DSL communication channel, link loss may be detected after 5 to 8 milliseconds; which is considerably smaller than legacy implementations, where the detection of a link loss may take up to 2 or 3 seconds.

Although the invention has been shown and described with respect to certain preferred embodiments, equivalents and modifications will occur to others skilled in the art upon the reading and understanding of the specification. The present invention includes all such equivalents and modifications and is limited only by the scope of the appended claims.

E.g., while above various examples have been disclosed in the context of DSL protocols, respective techniques may be readily applied to other kinds and types of communication systems.

The invention claimed is:

1. A method, comprising:

receiving a coded signal via a physical link,  
decoding the received coded signal to obtain a result signal,

detecting a link loss of a disturber physical link based on the coded signal, the decoding of the coded signal, or both, wherein an error correction code is determined to combat noise related to the disturber physical link, wherein an action is to be taken responsive to the link loss of the disturber physical link that includes a mitigation of an effect related to the link loss of the disturber physical link, the link loss being further determined based on energy across a plurality of samples in a frequency domain.

2. The method of claim 1,

wherein the link loss of the physical link is detected based on a temporal evolution of the at least one of the coded signal and/or the decoding in a time interval having a duration in the range of 3-15 ms.

3. The method of claim 1,

wherein the link loss is detected based on the decoding of the coded signal,

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wherein the decoding provides an error signal indicative of a presence of errors in the coded signal, wherein the link loss is detected based on the error signal.

4. The method of claim 3, wherein the error signal is indicative of a number of adjacent erroneous symbols in the coded signal, wherein the method further comprises:

- executing a threshold comparison between the number of adjacent erroneous symbols and a predefined threshold,

wherein the link loss is selectively detected based on a result of the threshold comparison.

5. The method of claim 3, wherein the decoding of the coded signal comprises decoding using a second-stage redundancy decoder operating based on the error signal.

6. The method of claim 1, wherein the decoding of the coded signal comprises decoding using a Viterbi decoder.

7. The method of claim 6, wherein the link loss is detected based on the decoding of the coded signal, wherein the decoding provides an error signal indicative of a presence of errors in the coded signal, wherein the link loss is detected based on the error signal, wherein the method further comprises

- determining the error signal based on a difference between different Viterbi paths of the Viterbi decoder.

8. The method of claim 1, wherein the decoding of the coded signal comprises decoding using a Quadrature Amplitude Modulation, QAM, decoder.

9. The method of claim 8, wherein the link loss is detected based on the decoding of the coded signal, wherein the decoding provides an error signal indicative of a presence of errors in the coded signal, wherein the link loss is detected based on the error signal, wherein the method further comprises:

- determining the error signal based on an Euclidean distance between a constellation point of a symbol of the coded signal and a constellation point of a predefined carrier.

10. The method of claim 1, wherein the decoding of the coded signal comprises decoding using a low-density parity check, LDPC, decoder.

11. The method of claim 10, wherein the link loss is detected based on the decoding of the coded signal, wherein the decoding provides an error signal indicative of a presence of errors in the coded signal, wherein the link loss is detected based on the error signal, wherein the method further comprises

- determining the error signal based on a log-likelihood estimate between a constellation point of a symbol of the coded signal and a constellation point of a predefined carrier.

12. The method claim 1, wherein the coded signal is modulated using at least one carrier,

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wherein the link loss is detected based on the coded signal, wherein the method further comprises:

- measuring energy levels of a plurality of resource blocks of the coded signal,

wherein the link loss is determined based on the measured energy levels.

13. The method of claim 12, further comprising:

- executing a threshold comparison between energy levels of the plurality of resource blocks and a predetermined threshold,

wherein the link loss is selectively determined based on a result of the threshold comparison.

14. The method of claim 12, wherein the coded signal is modulated using a plurality of carriers, wherein the plurality of resources blocks are adjacent time samples of the coded signal or adjacent carriers of the coded signal.

15. The method of claim 1, wherein the result signal comprises a sequence of bits forming transmission frames, wherein the link loss of the physical link is further detected based on a checksum of at least one of the transmission frames.

16. The method of claim 1, wherein the physical link comprises a copper wire, wherein the method further comprises:

- in response to determining the link loss of the physical link, removing a Digital Subscriber Line channel implemented via the physical link from a Digital Subscriber Line vector engine calculation, the Digital Subscriber Line vector engine calculation removing far-end crosstalk between the physical link and a plurality of further physical links.

17. The method of claim 16, further comprising:

- retaining further Digital Subscriber Line channels implemented via the further physical links at the Digital Subscriber Line vector engine calculation.

18. The method of claim 1, wherein the link loss corresponds to the physical link being interrupted and prohibiting communication.

19. The method of claim 1, wherein the action to be taken responsive to the link loss of the disturber physical link includes an action for a second physical link that is affected by the link loss of the disturber physical link.

20. A device, comprising:

- a receiver configured to receive a coded signal via a physical link, the coded signal having been coded by a Trellis coded modulation (TCM) encoder,
- a decoder configured to decode the received coded signal to obtain a result signal,
- at least one processor configured to detect a link loss of a disturber physical link based on the coded signal, the decoding of the coded signal, or both, wherein an error correction code is determined to combat noise related to the disturber physical link, wherein an action is to be taken responsive to the link loss of the disturber physical link that includes a mitigation of an effect related to the link loss of the disturber physical link, the link loss being further determined based on energy across a plurality of samples in a frequency domain.

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